

Topic 19C: Chromatography
6. know that chromatography separates components of a mixture between a mobile phase and a stationary phase
7. be able to calculate R_f values from one-way chromatograms
8. know that high performance liquid chromatography, HPLC, and gas chromatography, GC: - i are types of column chromatography - ii separate substances because of different retention times in the column - iii may be used in conjunction with mass spectrometry, in applications such as forensics or drugs testing in sport

Science Practical Endorsement

Overview

The assessment of practical skills is a compulsory requirement of the course of study for A level qualifications in biology, chemistry and physics. It will appear on all students' certificates as a separately reported result, alongside the overall grade for the qualification. The arrangements for the assessment of practical skills will be common to all awarding organisations. These arrangements include:

- A minimum of 12 practical activities to be carried out by each student which, together, meet the requirements of *Appendices 5b* (Practical skills identified for direct assessment and developed through teaching and learning) and *5c* (Use of apparatus and techniques) from the prescribed subject content, published by the Department for Education. The required practical activities will be defined by each awarding organisation.
- Teachers will assess students against Common Practical Assessment Criteria (CPAC) issued by the awarding organisations. The CPAC are based on the requirements of *Appendices 5b* and *5c* of the subject content requirements published by the Department for Education, and define the minimum standard required for the achievement of a pass.
- Each student will keep an appropriate record of their practical work, including their assessed practical activities
- Students who demonstrate the required standard across all the requirements of the CPAC will receive a 'pass' grade
- There will be no separate assessment of practical skills for AS qualifications.
- Students will answer questions in the AS and A level examination papers that assess the requirements of *Appendix 5a* (Practical skills identified for indirect assessment and developed through teaching and learning) from the prescribed subject content, published by the Department for Education. These questions may draw on, or range beyond, the practical activities included in the specification.

Setting practical work

Teaching and learning

Teachers should ensure that the core practicals listed in the subject content are incorporated into teaching and learning and that students carry them out. This is to support development of competency for the practical endorsement and also because students will be indirectly assessed on their practical skills in the examinations. Teachers should consider setting additional practical work to support teaching and learning.

Teachers must devise and retain a teaching plan that shows when practicals will be covered in teaching and learning.

Students should attempt all 16 practicals given in the qualification content.

Conditions for assessing practical work

Authenticity

Students and teachers must sign the practical competency authentication sheet (see *Appendix 4*).

Collaboration

Students may work in pairs on some practicals where appropriate, provided they are able to produce individual evidence to meet the competency statements.

Feedback

Teachers may help students to understand instructions and may provide feedback to support development of the competencies. Teachers should be confident that students have developed their skills and thus students should be able to demonstrate the competencies independently by the end of the course.

Resources

Students must have access to the equipment needed to carry out the core practicals. Students must also have access to IT and internet facilities.

Supervision

Teachers must ensure that students are supervised appropriately.

Evidence of practical work

Evidence should be collected of practical work that is sufficient to show that the competencies have been achieved. Evidence may take a variety of forms.

The practical activities prescribed in the specification provide opportunities for demonstrating competence in all the skills identified, together with the use of apparatus and techniques for each subject. However, students can also demonstrate these competencies in any additional practical activity undertaken throughout the course of study which covers the requirements of *Appendix 5c*.

Observations and notes by teachers

Teachers should observe sufficient student practicals to ensure that competencies have been achieved.

Student records

Students should keep notes of their practical work sufficient to show evidence of the practical competencies on making and recording observations and researching, referencing and reporting.

Assessing practical work

Teachers should make a judgement of student practical competence using the Common Practical Assessment Criteria on the following pages. Teachers should include any comments on the practical competency authentication sheet (see *Appendix 4*) to justify the decision made.

In order to achieve a pass, students will need to:

- develop these competencies by carrying out a minimum of 12 practical activities, which allow acquisition of the techniques outlined in the requirements of the specification
- consistently and routinely exhibit the competencies listed in the CPAC before the completion of the A-level course
- keep an appropriate record of their practical work, including their assessed practical activities
- be able to demonstrate and/or record independent evidence of their competency, including evidence of independent application of investigative approaches and methods to practical work.

Common Practical Assessment Criteria

Teachers must assess student practicals against the following competencies.

Criteria for the assessment of GCE Science practical competency for biology, chemistry and physics	
Competency	The criteria for a pass In order to be awarded a Pass a learner must, by the end of the practical science assessment, consistently and routinely meet the criteria in respect of each competency listed below. A learner may demonstrate the competencies in any practical activity undertaken as part of that assessment throughout the course of study. Learners may undertake practical activities in groups. However, the evidence generated by each learner must demonstrate that he or she independently meets the criteria outlined below in respect of each competency. Such evidence: (a) will comprise both the learner's performance during each practical activity and his or her contemporaneous record of the work that he or she has undertaken during that activity, and (b) must include evidence of independent application of investigative approaches and methods to practical work.
1. Follows written procedures	a) Correctly follows written instructions to carry out the experimental techniques or procedures.
2. Applies investigative approaches and methods when using instruments and equipment	a) Correctly uses appropriate instrumentation, apparatus and materials (including ICT) to carry out investigative activities, experimental techniques and procedures with minimal assistance or prompting. b) Carries out techniques or procedures methodically, in sequence and in combination, identifying practical issues and making adjustments when necessary. c) Identifies and controls significant quantitative variables where applicable, and plans approaches to take account of variables that cannot readily be controlled. d) Selects appropriate equipment and measurement strategies in order to ensure suitably accurate results.

Criteria for the assessment of GCE Science practical competency for biology, chemistry and physics	
3. Safely uses a range of practical equipment and materials	a) Identifies hazards and assesses risks associated with these hazards, making safety adjustments as necessary, when carrying out experimental techniques and procedures in the lab or field. b) Uses appropriate safety equipment and approaches to minimise risks with minimal prompting.
4. Makes and records observations	a) Makes accurate observations relevant to the experimental or investigative procedure. b) Obtains accurate, precise and sufficient data for experimental and investigative procedures and records this methodically using appropriate units and conventions.
5. Researches, references and reports	a) Uses appropriate software and/or tools to process data, carry out research and report findings. b) Cites sources of information demonstrating that research has taken place, supporting planning and conclusions.

Marking and standardisation

The practical work is assessed by teachers. Pearson will support teachers in making judgements against the criteria for assessment.

In coordination with other exam boards, Pearson will monitor how schools provide students with opportunities to develop and demonstrate the required practical skills and how they mark the assessments.

Every school will be monitored at least once in a two-year period in respect of at least one of the A level science subjects. These monitoring visits will be coordinated by JCQ, who will undertake communications with centres to facilitate the allocation of exam board monitoring visits.

In common with other exam boards, Pearson will require centres to provide a statement confirming they have taken reasonable steps to secure that students:

- undertook the minimum number of practical activities, and
- made a contemporaneous record of their work.

If a school fails to provide a statement, or provides a false statement, this will be treated as malpractice and/or maladministration.

Students will only get a certificate for the practical assessment if they achieve at least a grade E in the examined part of the qualification.

Students who do not pass the practical assessment will have a 'Not Classified' outcome included on their certificate unless they were exempt from the assessment because of a disability.

Malpractice

Candidate malpractice

Candidate malpractice refers to any act by a candidate that compromises or seeks to compromise the process of assessment or which undermines the integrity of the qualifications or the validity of results/certificates.

Candidate malpractice in controlled assessments discovered before the candidate has signed the declaration of authentication form does not need to be reported to Pearson.

Candidate malpractice found in controlled assessments after the declaration of authenticity has been signed, and in examinations **must** be reported to Pearson on a *JCQ M1 Form* (available at www.jcq.org.uk/exams-office/malpractice). The completed form can be emailed to pqsmalpractice@pearson.com or posted to Investigations Team, Pearson, 190 High Holborn, London, WC1V 7BH. Please provide as much information and supporting documentation as possible. Note that the final decision regarding appropriate sanctions lies with Pearson.

Failure to report candidate malpractice constitutes staff or centre malpractice.

Staff/centre malpractice

Staff and centre malpractice includes both deliberate malpractice and maladministration of our qualifications. As with candidate malpractice, staff and centre malpractice is any act that compromises or seeks to compromise the process of assessment or undermines the integrity of the qualifications or the validity of results/certificates.

All cases of suspected staff malpractice and maladministration **must** be reported immediately, before any investigation is undertaken by the centre, to Pearson on a *JCQ M2(a) Form* (available at www.jcq.org.uk/exams-office/malpractice). The form, supporting documentation and as much information as possible can be emailed to pqsmalpractice@pearson.com or posted to Investigations Team, Pearson, 190 High Holborn, London, WC1V 7BH. Note that the final decision regarding appropriate sanctions lies with Pearson.

Failure to report malpractice itself constitutes malpractice.

More-detailed guidance on malpractice can be found in the latest version of the document *JCQ General and Vocational Qualifications Suspected Malpractice in Examinations and Assessments*, available at www.jcq.org.uk/exams-office/malpractice.

Assessment

Assessment summary

Students must complete all assessment in May/June in any single year.